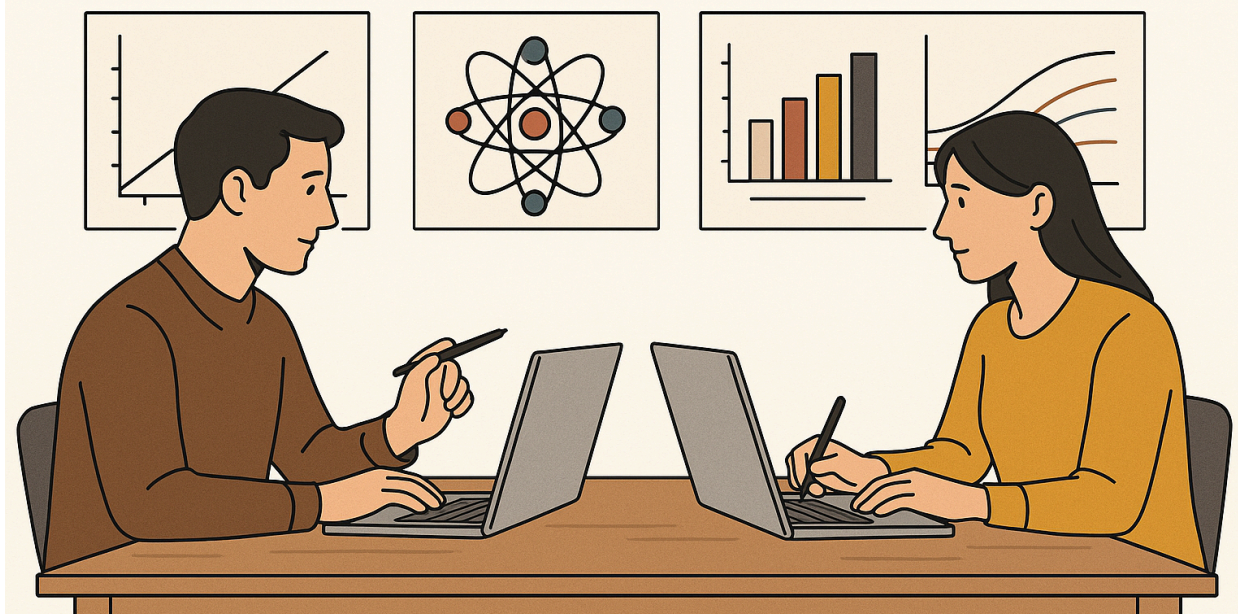


Welcome to the ALD/E-ImageMiner Pilot Annotation Task



ALD/E-ImageMiner Pilot Annotation Task – Guidelines

Welcome to the **ALD/E-ImageMiner Pilot Annotation Task**. We are glad you signed up and hope you find this experience enriching, fun, and stimulating to your knowledge as well. This **pilot task** will help us screen annotators that we will hire for the full annotation project.

Overview

This pilot phase consists of **10 scientific articles** on the topics of atomic layer deposition (ALD) and atomic layer etching (ALE), situated within the broader field of materials science and engineering. Your annotation task focuses on the **figures** extracted from each of these articles. You can browse the collection of pilot papers in our GitHub repository here:

👉 [Pilot Annotation Task Papers](#)

The work will be carried out in a **personal Google Sheet workbook** that we provide. Each workbook contains **10 sheets**, with each sheet corresponding to one paper. In this way, the rows of each sheet represent the individual figures from the assigned article.

The overall goal of this pilot is to **curate and correct figure annotations** that were originally generated by the Qwen-2.5-VL-7B model. In some cases, the model annotations may already be correct, requiring no further action from you. In other cases, the model may be partially or completely wrong, and you will need to correct or recreate the annotation. For simple cases, we expect that the auto-generated annotations will already be accurate, saving you time. The real value of this project lies in the **human-corrected annotations** where the model falls short.

To help you understand the process more concretely, we also provide a **running example annotation sheet** that you can consult at any time: [Example Annotation Sheet – Annotator 1](#). This example illustrates how each column (with the exception of column I) is expected to be filled out and can serve as a reference alongside these written guidelines.

Using the Workbook

When you begin working on a sheet (one paper), please first write your **Prolific ID** in **Column A, Row 2**. You only need to enter it once per sheet, but since there are 10 papers, you will enter it **10 times in total** across the entire workbook.

Column B will provide a link to the **full paper PDF** in our GitHub repository. Before beginning annotation, we encourage you to briefly skim the paper to understand its figures in context. (See the example sheet).

The rows in **Column C** will contain links to the **extracted figure image** stored in the repository. These are the same figures taken directly from the original PDF and saved as separate images. **Column D** shows the **figure label** from the paper (e.g., “Figure 3”).

This setup keeps the annotation tool simple and lightweight, while still giving you easy access to the papers and figures you will be working on.

The Four Annotation Tasks

You may not need to do all four tasks for every figure — the rules below explain when to stop.

1. Figure Classification

- This task relies on a **taxonomy of 40 figure types**, including an “*unknown*” category.
 - See the [full list here](#), which includes brief descriptions, parent taxonomy, and alternative names.
- A figure may be composed of **multiple subfigures**.
 - Read left to right, top to bottom, and follow the *a, b, c ...* labeling scheme if subfigures are not already labeled.
- Steps:
 - Open the link in **Column E (auto class)** and verify if it is correct.
 - If correct → leave it.
 - If wrong → enter the correct annotation(s) in **Column F (manual class)** and briefly explain why in **Column G (comment auto class)**.
 - *Reminder*: a label and a class are tab-separated (see the example sheet for reference).
- **Next step rule:**
 - Only continue with Tasks 2–4 if the figure (or **all subfigures**) belongs to the parent taxonomy “**quantitative-plot**” or “**quantitative-plot (domain-specific)**”.
 - If not → skip to the next row.
 - If the figure contains mixed subfigures (some quantitative, some not) → skip to the next row.
- 💡 *Tip*: You may use ChatGPT to assist with classification or justification (e.g., upload the figure and taxonomy and ask for candidate labels).
 - However, the **final decision is yours**. AI tools can only help check your reasoning.
 - Using AI to *perform* the task defeats the purpose of this project.

2. Data Table Curation

- This annotation project includes only **quantitative plots**, and every such plot is based on underlying numeric data.
- The goal of this task is to curate that numeric data in a usable form.
- In your workbook sheet, you will open the link in **Column H (“auto data”)** to view the auto-extracted data.
 - In many cases, this auto data will be a good starting point.
- Your steps:
 - Copy the content from **Column H** into **Column I (manual data)**.
 - Correct any errors or inconsistencies.
- Guidelines:
 - **Do not remove newlines** → each data point should remain on a separate line.
 - You do **not** need to annotate every single axis point.
 - Instead, capture only the **salient points** that help minimally reconstruct the plot.
- Example: Subfigure data annotations should look like this: [figure_9.data.txt](#)

3. Figure Content Summary Curation

- In your workbook sheet, open the link in **Column J (“auto summary”)**.
 - The file contains a summary generated by Qwen about the content of the figure or its subfigures.
- Your task is to validate this summary yourself.
- In many cases, Qwen’s summaries are **too vague, generic, or repetitive** and do not fully capture the figure content.
- If the auto summary is acceptable → leave it as is.
- If it is not acceptable →
 - Write the corrected version in **Column K (“manual summary”)**.
 - Provide a brief justification in **Column L (“comment on auto summary”)** explaining why the auto version was wrong.
- Refer to the [example sheet](#) (especially rows with quantitative plots) for how to format entries across **Columns J, K, and L**.

4. Figure Q&A Annotation

- This task requires you to create **4 question–answer (Q&A) pairs per figure**.
Question types (choose one per question):
 1. Process-Oriented
 2. Comparative/Trend
 3. Structure–Property
 4. Application/Performance
Answer types (choose one per answer):
 1. Yes/No
 2. Factoid
 3. List
 4. Paragraph (≥3 sentences)
- For detailed definitions and examples of these types, we ask you to read the following documentation: [Q&A README](#)
- Please read that file carefully before starting, and keep it at hand while annotating.
- **In your workbook sheet:**
 - For the 1st Q&A:
 - **Column M (Q1 type)** → select the question type.
 - **Column N (Q1)** → write your question.
 - **Column O (A1 type)** → select the answer type.
 - **Column P (A1)** → write your answer.
 - Repeat for Q2 (Columns Q–T), Q3 (Columns U–X), and Q4 (Columns Y–AB).
- **Guidelines:**
 1. Each figure must have **4 Q&As**.
 2. At least **two-thirds of figures in a paper** must include at least one **paragraph-style answer** (≥3 sentences).
 - Example: For 6 figures, at least 4 must include paragraph answers.
 3. No more than **one-third of answers** in a paper may be **yes/no style**.
 - Example: For 6 figures, at most 2 may be yes/no.
 4. Avoid skewing: use a **balanced mix** of question and answer types.
 5. You are free to choose combinations of question and answer types, as long as these balance rules are met.

AI use:

-  **Do not use AI to generate questions or answers.** Both must be your own original work.
 - Part of the annotator compensation is specifically for the time and effort required to write original questions and answers based on your own ideas, creativity, expertise, and impressions from the images.
 - Our team is already well equipped to run vision–language models (VLMs) to generate Q&As, so this task has a *different purpose*: to capture authentic **human annotations** that cannot simply be reproduced by machines.
-  **You may use AI in a minimally non-invasive way only.**
 - Acceptable uses include double-checking linguistic aspects such as grammar, spelling, verbosity, or style.
 - AI should act only as a language helper, never as the source of scientific content or reasoning.

👉 The value of this project lies in your **human judgment, creativity, and unique perspective** when interpreting the images — not in outputs that machines could already generate.

Completion, Next Steps, and Contact

Annotation **quality** is the main criterion for selecting annotators after the pilot phase. We will evaluate your work based on careful attention to the instructions, correctness of classifications, curated data, summaries, and Q&As, as well as adherence to the balance rules (e.g., paragraph requirements and limits on yes/no answers). In short, we are looking for thoughtful, consistent, and reliable annotations that show care and understanding of the figures.

All annotators will be **compensated upon completing the pilot task** (10 papers). After the pilot, a group of selected annotators will continue with the **full annotation task** (20 papers). Of these, **5 papers** will be repeated from the pilot and shared across all annotators to measure **inter-annotator agreement**, while the other **15 papers** will be unique to each annotator. We plan to hire **20 annotators** for this phase.

The long-term goal is to build a dataset of **300 fully annotated papers**, creating a valuable research resource for the community. By participating, you are directly contributing to advancing machine understanding of scientific images, with an impact on future AI and materials science research.

Once you have read these instructions carefully, please email us directly to request your annotation sheet for the pilot task:

- **fahad.ahmed [at] tib.eu**
- **jennifer.dsouza [at] tib.eu**

We will respond with your personalized Google Sheet workbook so you can begin annotating. We look forward to working with you and deeply value your contribution to this exciting project.

THANK YOU!

